

Maximum Likelihood Estimation in Greater Depth

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Parameter Estimation

- In many applications, we have samples of a random variable which we know belongs to a distribution of so and so family (eg: Poisson, normal, etc.)
- However, we may **not** know *some* or *any* of the parameters of the distribution.
- We can however **estimate** the parameters given some *assumptions* about the samples. This is called **parameter estimation**.

Parameter Estimation

- Let X be a continuous random variable with probability density $f_X(x;\theta)$.
- Consider a sample of X having value say x_1 . Then we say that the **likelihood** of this sample is $f_X(x_1;\theta)$ assuming parameter θ .
- If we observe some n samples (rather n samples, one from each of the n iid random variables having density $f_X(x;\theta)$), then their **joint likelihood** is given as $f_{x_1, x_2, x_3, \dots, x_n}(x_1, x_2, \dots, x_n; \theta)$ or in shorthand $f(x_1, x_2, \dots, x_n; \theta)$.

Parameter Estimation

- Now consider the case where we have the n sample values but we do not know θ .
- A reasonable way to estimate θ is to treat the joint likelihood as a function of θ and maximize it with respect to θ .
- Intuitively, such an estimate is the one which maximizes the probability density of the observed values. It is called as the **maximum likelihood estimate** of θ .

Parameter estimation

- 1) Maximum likelihood estimation is **equally applicable for discrete random variables** in which case the likelihood is defined in terms of discrete probabilities instead of probability density functions.
- 2) In maximum likelihood estimation, it is often easier (in terms of calculations) to maximize $\log f(x_1, x_2, \dots, x_n; \theta)$ instead of $f(x_1, x_2, \dots, x_n; \theta)$.
- 3) Doing so does not alter the estimate in any way for many densities that are not zero-valued except at $\pm$ infinity.

Regularity Conditions

- In all our derivations for MLE, we made some assumptions implicitly.
- We will state them here explicitly:
 - R1: All CDFs are distinct, that is $\theta \neq \theta' \rightarrow F(x;\theta) \neq F(x;\theta')$
 - R2: All PDFs have the same *support* (i.e., regions where the PDF is non-zero) for different values of θ . In other words, the PDF support should not depend on θ .
 - R3: The parameter θ of interest lies in the interior of the domain on which the PDF is defined.

MLE: A deeper look

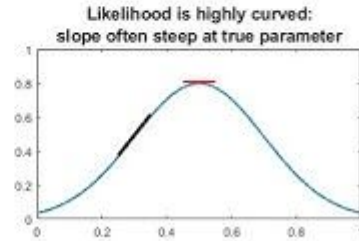
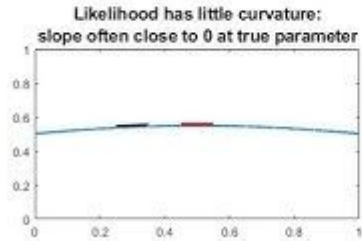
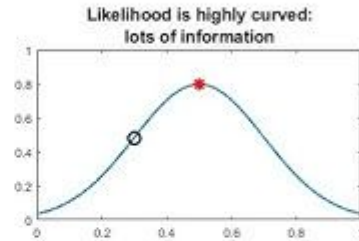
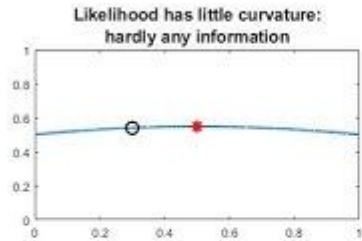
- Efficiency of the MLE
- Functional invariance of the MLE

Efficiency of the MLE

- We will state and prove the **Cramer-Rao bound**, the *lower bound* on the **variance** of *any* unbiased estimator.
- We will then show that the **MLE achieves this bound asymptotically**.
- An unbiased estimator that achieves this lower bound is said to be efficient; the MLE is thus asymptotically efficient.
- For developing these bounds, we will assume two more regularity conditions:
 - R4: The PDF is twice differentiable w.r.t. the parameter θ .
 - R5: The integral $\int f(x;\theta) dx$ can be differentiated twice under the integral sign w.r.t. the parameter θ .
- For developing these bounds, we will need to first develop the concept of **Fisher information**.

Concept of Fisher Information

Slopes



X axis: parameter θ
Y axis: log likelihood i.e.
 $\log f(x;\theta)$ at some value of x

https://www.youtube.com/watch?v=f_wU0LeNUvE

<https://www.youtube.com/watch?v=82molmnRCg0>

Concept of Fisher Information

$$1 = \int_{-\infty}^{\infty} f(x; \theta) dx.$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial f(x; \theta)}{\partial \theta} dx.$$

Taking derivatives w.r.t. θ

$$0 = \int_{-\infty}^{\infty} \frac{\partial f(x; \theta) / \partial \theta}{f(x; \theta)} f(x; \theta) dx$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx \quad \longrightarrow \quad E \left[\frac{\partial \log f(X; \theta)}{\partial \theta} \right] = 0;$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx \quad \longrightarrow \quad E \left[\frac{\partial \log f(X; \theta)}{\partial \theta} \right] = 0;$$

Differentiate both sides w.r.t. θ

$$0 = \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx + \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx.$$

$$I(\theta) = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx = E \left[\left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)^2 \right]$$

This term is called the **Fisher information**. It is always non-negative. It is named after Sir Ronald Fisher, a legendary statistician.



Fisher–Race blood group system
Behrens–Fisher problem
Cornish–Fisher expansion
von Mises–Fisher distribution
family allowance
Wright–Fisher model
Ancillary statistic
Fiducial inference
Intraclass correlation
Infinitesimal model
Inverse probability
Null hypothesis
Maximum likelihood estimation
Neutral theory of molecular evolution
Particulate inheritance
p-value
Random effects model
Relative species abundance
Reproductive value
Sufficient statistic
Analysis of variance
Variance

Fisher's exact test
Fisher's inequality
Fisher's principle
Fisher's geometric model
Fisher's Iris data set
Fisher's linear discriminant
Fisher's equation
Fisher information
Fisher's method
Fisherian runaway
Fisher's fundamental theorem of natural selection
Fisher's noncentral hypergeometric distribution
Fisher's z-distribution
Fisher transformation
Fisher consistency
F-distribution
F-test
Fisher–Tippett distribution
Fisher–Tippett–Gnedenko theorem
Fisher–Yates shuffle

https://en.wikipedia.org/wiki/Ronald_Fisher

PhD advisor of C. R. Rao

$$I(\theta) = \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx = E \left[\left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)^2 \right]$$

$$0 = \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx + \int_{-\infty}^{\infty} \frac{\partial \log f(x; \theta)}{\partial \theta} \frac{\partial \log f(x; \theta)}{\partial \theta} f(x; \theta) dx.$$

$$I(\theta) = - \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx = -E \left[\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2} \right]$$

Score
function

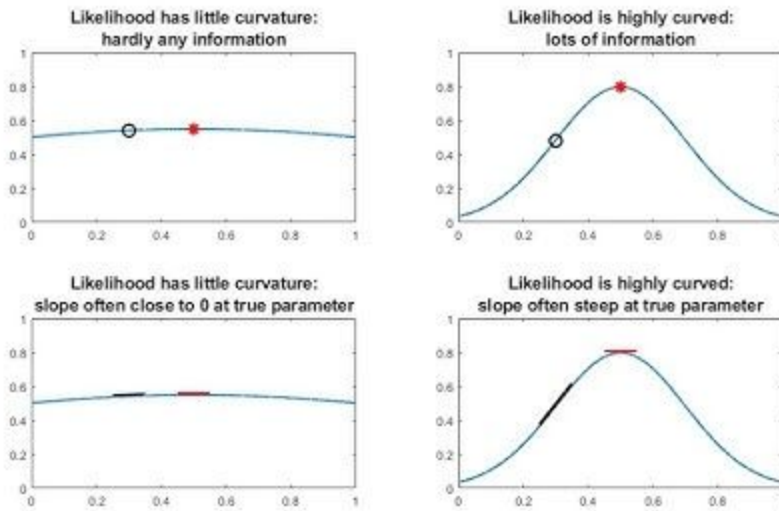
$$E \left[\frac{\partial \log f(X; \theta)}{\partial \theta} \right] = 0 \quad \longrightarrow \quad I(\theta) = \text{Var} \left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)$$

The Fisher information is the variance of the score function.

Why is this called the Fisher **information**?

- It is a way to measure the amount of information that an observed random variable X (observed via its **samples**) carries about the parameter θ which characterizes the PDF/PMF $f_X(\cdot; \theta)$ of X .
- If f_X (or its logarithm) is **sharply peaked** around a given value of θ , we say it provides a lot of information regarding θ , **as any change in θ causes large changes in f_X (or its logarithm).**
- If f_X is flat or spread out around a given value of θ , **we would need too many samples to estimate θ reliably.**
- See figure on the next slide.

Slopes



X axis: parameter θ
Y axis: log likelihood i.e.
 $\log f(x;\theta)$ at some value of x

https://www.youtube.com/watch?v=f_wU0LeNUvE

<https://www.youtube.com/watch?v=82molmnRCg0>

Why is this called the Fisher **information**?

- We have previously argued that the Fisher information is the variance of the score.
- The **score function** is the derivative of the log likelihood w.r.t. the parameter θ .
- A random variable with high Fisher information implies that the **absolute value of the score** of that random variable is often high.

$$I(\theta) = - \int_{-\infty}^{\infty} \frac{\partial^2 \log f(x; \theta)}{\partial \theta^2} f(x; \theta) dx = -E \left[\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2} \right]$$

$$I(\theta) = \text{Var} \left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)$$

Why is this called the Fisher **information**?

- The Fisher information is also the negative of the expected value of the second derivative (**curvature**) of the log likelihood function with respect to θ .
- Near the true value of a parameter, a low Fisher information indicates that the log-likelihood function changes very slowly with respect to θ .
- Near the true value of a parameter, a high value of Fisher information indicates that the log-likelihood function changes very rapidly with respect to θ .
- We will now work out the Fisher information for some simple univariate distributions.

Fisher Information for mean of a Univariate Gaussian with fixed σ

$$f(x; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$\ell(\mu; x) = \log f(x; \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}$$

$$\frac{\partial \ell(\mu; x)}{\partial \mu} = \frac{(x - \mu)}{\sigma^2}.$$

$$I(\mu) = \mathbb{E}\left[\left(\frac{\partial \ell(\mu; X)}{\partial \mu}\right)^2\right] \quad I(\mu) = \mathbb{E}\left[\frac{(X - \mu)^2}{\sigma^4}\right] = \frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2]$$

$$I(\mu) = \frac{1}{\sigma^2}$$

The Fisher information of a Gaussian with low variance is more!

Fisher Information for mean of a Univariate Gaussian with fixed σ

$$I_n(\mu) = nI(\mu) = \boxed{\frac{n}{\sigma^2}}.$$

The Fisher information of a Gaussian with low variance is more!

If we had n independent samples of the Gaussian.

Fisher Information for variance of a Univariate Gaussian with fixed mean

$$f(x; v) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x - \mu)^2}{2v}\right)$$

$$\ell(v; x) = -\frac{1}{2} \log(2\pi v) - \frac{(x - \mu)^2}{2v}.$$

$$\frac{\partial \ell(v; x)}{\partial v} = -\frac{1}{2v} + \frac{(x - \mu)^2}{2v^2} = \frac{(x - \mu)^2 - v}{2v^2}$$

$$I(v) = \mathbb{E}\left[\left(\frac{\partial \ell(v; X)}{\partial v}\right)^2\right] = \mathbb{E}\left[\left(\frac{(X - \mu)^2 - v}{2v^2}\right)^2\right] = \frac{1}{4v^4} \mathbb{E}\left[\left((X - \mu)^2 - v\right)^2\right]$$

Fisher Information for variance of a Univariate Gaussian with fixed mean

$$I(v) = \mathbb{E} \left[\left(\frac{\partial \ell(v; X)}{\partial v} \right)^2 \right] = \mathbb{E} \left[\left(\frac{(X - \mu)^2 - v}{2v^2} \right)^2 \right] = \frac{1}{4v^4} \mathbb{E} \left[((X - \mu)^2 - v)^2 \right]$$

$$\mathbb{E}[(X - \mu)^2] = v, \quad \text{and} \quad \mathbb{E}[(X - \mu)^4] = 3v^2$$

$$\mathbb{E} \left[((X - \mu)^2 - v)^2 \right] = \mathbb{E}[(X - \mu)^4] - 2v \mathbb{E}[(X - \mu)^2] + v^2 = 3v^2 - 2v^2 + v^2 = 2v^2$$

$$I(v) = \frac{1}{4v^4} \cdot 2v^2 = \boxed{\frac{1}{2v^2}}$$

Notice again: lower variance implies higher Fisher information!

$$I_n(v) = nI(v) = \boxed{\frac{n}{2v^2}}$$

With n samples

Fisher information of a Bernoulli distribution

$$\Pr(X = 1) = p, \quad \Pr(X = 0) = 1 - p$$

$$f(x; p) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}$$

$$\ell(p; x) = x \log p + (1 - x) \log(1 - p)$$

$$\frac{\partial \ell(p; x)}{\partial p} = \frac{x}{p} - \frac{1 - x}{1 - p} = \frac{x - p}{p(1 - p)}$$

$$I(p) = \mathbb{E} \left[\left(\frac{\partial \ell(p; X)}{\partial p} \right)^2 \right]$$

Fisher information of a Bernoulli distribution

$$I(p) = \mathbb{E} \left[\left(\frac{\partial \ell(p; X)}{\partial p} \right)^2 \right]$$

$$I(p) = \mathbb{E} \left[\left(\frac{X - p}{p(1 - p)} \right)^2 \right] = \frac{1}{p^2(1 - p)^2} \mathbb{E}[(X - p)^2]$$

$$\text{Var}(X) = \mathbb{E}[(X - p)^2] = p(1 - p)$$

$$I(p) = \frac{1}{p(1 - p)} \qquad I_n(p) = nI(p) = \boxed{\frac{n}{p(1 - p)}}$$

Fisher Information Matrix

- If there are many parameters in the PDF (Eg: Gaussian), then a scalar-valued Fisher information is replaced by a **Fisher information matrix**.
- This is defined as follows:

$$f_X(x; \boldsymbol{\theta}), \boldsymbol{\theta} := (\theta_1, \theta_2, \dots, \theta_k)$$
$$[\mathcal{I}(\boldsymbol{\theta})]_{ij} = E \left[\frac{\partial \log f_X(x; \boldsymbol{\theta})}{\partial \theta_i} \frac{\partial \log f_X(x; \boldsymbol{\theta})}{\partial \theta_j} \right]$$

This is a $k \times k$ positive-semi-definite matrix (for k parameters). We will not study the Fisher information matrix further.

Cramer-Rao Lower Bound Theorem

- Let $X_1, X_2, \dots, X_n$ be iid samples from $f(x; \theta)$.
- Assume that the regularity conditions R1 to R5 hold.
- Let $Y = u(X_1, X_2, \dots, X_n)$ be an estimator of θ with mean $E(Y) = k(\theta)$.
- Then the variance of Y is lower-bounded as follows:

$$\text{Var}(Y) \geq \frac{[k'(\theta)]^2}{nI(\theta)}.$$


Fisher information

This result is called the **Cramer-Rao lower-bound theorem** or just the **Cramer-Rao lower bound** (CRLB). It is named after Harald Cramer and C. R. Rao.



Cramér–Rao bound

Cramér's conjecture

Cramér's decomposition theorem

Cramér's theorem (large deviations)

Cramér–Wold theorem

Cramér–von Mises criterion

Cramér function of large deviations
theory

Ruin theory

Cramér's V

Cramér's inequality

https://en.wikipedia.org/wiki/Harald_Cram%C3%A9r



Multivariate statistical theory

Cramér–Rao bound

Rao–Blackwell theorem

Orthogonal arrays

Score test

[https://en.wikipedia.org/wiki/C. R. Rao](https://en.wikipedia.org/wiki/C._R._Rao)

Calyampudi Radhakrishna Rao

Proof of Cramer-Rao Lower bound

By definition of expectation and independence of samples

$$k(\theta) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, \dots, x_n) f(x_1; \theta) \cdots f(x_n; \theta) dx_1 \cdots dx_n$$

By taking derivatives w.r.t. θ

$$\begin{aligned} k'(\theta) = & \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, x_2, \dots, x_n) \left[\sum_1^n \frac{1}{f(x_i; \theta)} \frac{\partial f(x_i; \theta)}{\partial \theta} \right] \\ & \times f(x_1; \theta) \cdots f(x_n; \theta) dx_1 \cdots dx_n \end{aligned}$$

$$k'(\theta) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, x_2, \dots, x_n) \left[\sum_1^n \frac{\partial \log f(x_i; \theta)}{\partial \theta} \right] \\ \times f(x_1; \theta) \cdots f(x_n; \theta) dx_1 \cdots dx_n.$$

$$Z = \sum_1^n [\partial \log f(X_i; \theta) / \partial \theta]$$

$$E(Z) = 0$$

We proved this on slide 13.

$$\text{Var}(Z) = nI(\theta)$$

Again on slide 13 (but there we had $n = 1$).

$$k'(\theta) = E(YZ)$$

$$k'(\theta) = E(YZ)$$

$$k'(\theta) = E(YZ) = E(Y)E(Z) + \rho\sigma_Y\sqrt{nI(\theta)}$$

Std. dev. of Y

Correlation coefficient
between Y and Z

$$\rho = \frac{k'(\theta)}{\sigma_Y\sqrt{nI(\theta)}}$$

Because $E(Z) = 0$

$$\rho^2 \leq 1$$

$$\frac{[k'(\theta)]^2}{\sigma_Y^2 n I(\theta)} \leq 1$$

$$\text{Var}(Y) \geq \frac{[k'(\theta)]^2}{nI(\theta)}.$$

Corollary to the Cramer-Rao Bound

- In the earlier statement if Y was an unbiased estimator of θ , then we have $k(\theta) = \theta$ and $k'(\theta) = 1$.
- The bound then reduces to the following:

$$\text{Var}(Y) \geq \frac{1}{nI(\theta)}$$

Significance of the Cramer-Rao Bound

- The bound states that the variance for any unbiased estimator is at least $1/nI(\theta)$.
- It turns out (we shall prove) that the maximum likelihood estimator (MLE) achieves this lower bound asymptotically.
- This means that no **unbiased** estimator can have a variance that is **lower** than of the **MLE**.
- Note that in some cases, a **biased estimator** may still produce an MSE that is **lower** than the Cramer-Rao lower bound.
- Also, in some cases, there may **not** exist an unbiased estimator that achieves the Cramer-Rao lower bound.
- An unbiased estimator Y of parameter θ is said to be an **efficient estimator** of θ if its variance equals the Cramer-Rao bound.
- The ratio of the Cramer-Rao lower bound to the actual variance of the unbiased estimator is called the **efficiency** of the estimator.

Example 1: Poisson distribution

- Consider samples $X_1, X_2, \dots, X_n$ from $\text{Poisson}(\theta)$.
- We know the MLE is the sample mean – it is an **unbiased** estimator.
- The variance of the MLE is θ/n .
- Now, we have:

$$\begin{aligned}\frac{\partial \log f(x; \theta)}{\partial \theta} &= \frac{\partial}{\partial \theta} (x \log \theta - \theta - \log x!) \\ &= \frac{x}{\theta} - 1 = \frac{x - \theta}{\theta}.\end{aligned}$$

$$E \left[\left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)^2 \right] = \frac{E(X - \theta)^2}{\theta^2} = \frac{\sigma^2}{\theta^2} = \frac{\theta}{\theta^2} = \frac{1}{\theta}$$

Example 1: Poisson distribution

- The Cramer-Rao lower bound is $1/n(1/\theta) = \theta/n$.
- As the MLE whose variance is θ/n , we see that the MLE here attains the Cramer-Rao bound.
- In this case, the MLE is an **efficient** estimator of θ .

Example 2: An unbiased variant of the MLE for variance of a Gaussian

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad \theta = \sigma^2$$

$$I_1(\theta) = \frac{1}{2\theta^2}, \quad I_n(\theta) = \frac{n}{2\theta^2}.$$

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)} = \frac{2\theta^2}{n} = \frac{2\sigma^4}{n}.$$

Fisher information

Cramer-Rao lower bound for any estimator of θ .

Example 2: An unbiased variant of the MLE for variance of a Gaussian

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

This is an estimator of the variance σ^2 . It is an unbiased estimator obtained by multiplying the MLE estimate with $n/(n-1)$.

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$$

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1} > \frac{2\sigma^4}{n} = \text{CRLB}$$

This is not an efficient estimator! Its **efficiency** is $(n-1)/n$. It is however **asymptotically efficient**.

The MLE is asymptotically efficient

- The efficiency of the previous estimator is *asymptotically* 1 (i.e. it is 1 when n tends to infinity).
- Let $X_1, X_2, \dots, X_n$ be iid samples from $f(x; \theta_0)$.
- Assume that the regularity conditions R1 to R5 hold.
- Additionally assume that the PDF is also thrice differentiable as a function of θ_0 such that

$$\left| \frac{\partial^3}{\partial \theta^3} \log f(x; \theta) \right| \leq M(x)$$

- Further assume that the Fisher information is positive and less than infinity.
- Then the MLE must satisfy the following:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N\left(0, \frac{1}{I(\theta_0)}\right)$$

The MLE is asymptotically efficient

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N\left(0, \frac{1}{I(\theta_0)}\right)$$

- This implies that the variance of the Maximum likelihood estimate ($\hat{\theta}$) of θ_0 has a variance that is asymptotically equal to $1/I(\theta_0)$.
- Moreover the MLE $\hat{\theta}$ is **Gaussian distributed** with a mean equal to the **true parameter** θ_0 and a variance given by the *reciprocal of the Fisher information*.
- We prove this result on the next slide, **but this proof is not on the exam**.

$$l'(\theta) = \frac{\partial \log f(x; \theta)}{\partial \theta}$$

True
parameter



$$l'(\hat{\theta}_n) = l'(\theta_0) + (\hat{\theta}_n - \theta_0)l''(\theta_0) + \frac{1}{2}(\hat{\theta}_n - \theta_0)^2 l'''(\theta_n^*)$$

θ_n^* is between θ_0 and $\hat{\theta}_n$ MLE



$$l'(\hat{\theta}_n) = 0$$

Derivative of the log-likelihood is zero at the maximum of the log-likelihood

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{n^{-1/2}l'(\theta_0)}{-n^{-1}l''(\theta_0) - (2n)^{-1}(\hat{\theta}_n - \theta_0)l'''(\theta_n^*)}$$

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{n^{-1/2}l'(\theta_0)}{-n^{-1}l''(\theta_0) - (2n)^{-1}(\hat{\theta}_n - \theta_0)l'''(\theta_n^*)}$$

$$\frac{1}{\sqrt{n}}l'(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(X_i; \theta_0)}{\partial \theta} \xrightarrow{D} N(0, I(\theta_0))$$

- This follows from the definition of $l'(\theta_0)$ and the central limit theorem.
- Why does the CLT apply?
 - Every term in the summation has an expected value of 0 (why? See slide 13)
 - The variance of every term in the summation is

$$\text{Var}(\partial \log f(X_i; \theta_0) / \partial \theta) = I(\theta_0) < \infty$$

$$-\frac{1}{n}l''(\theta_0) = -\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(X_i; \theta_0)}{\partial \theta^2} \xrightarrow{P} I(\theta_0)$$

By law of large numbers

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{n^{-1/2}l'(\theta_0)}{-n^{-1}l''(\theta_0) - (2n)^{-1}(\hat{\theta}_n - \theta_0)l'''(\theta_n^*)}$$

$$\hat{\theta}_n - \theta_0 \xrightarrow{P} 0$$

This quantity is bounded (i.e. less than infinity). Hence the second term in the denominator tends to 0.

$$\frac{1}{\sqrt{n}}l'(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(X_i; \theta_0)}{\partial \theta} \xrightarrow{D} N(0, I(\theta_0))$$

$$-\frac{1}{n}l''(\theta_0) = -\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(X_i; \theta_0)}{\partial \theta^2} \xrightarrow{P} I(\theta_0)$$

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N\left(0, \frac{1}{I(\theta_0)}\right)$$

The MLE is asymptotically efficient

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N\left(0, \frac{1}{I(\theta_0)}\right)$$

- This result has great practical significance.
- We can use the MLE $\hat{\theta}$ to construct approximate confidence interval for the parameter θ_0 .
- The asymptotic standard deviation of $\hat{\theta}$ is $[nI(\theta_0)]^{-1/2}$.
- We can use
$$I(\hat{\theta}_n) \xrightarrow{P} I(\theta_0)$$

This yields the following approximate confidence interval for θ_0

$$\left(\hat{\theta}_n - z_{\alpha/2} \frac{1}{\sqrt{nI(\hat{\theta}_n)}}, \hat{\theta}_n + z_{\alpha/2} \frac{1}{\sqrt{nI(\hat{\theta}_n)}} \right)$$

Functional Invariance of MLEs

- **Invariance theorem for MLEs:** If θ' is the MLE of the parameter θ of a density function $f_{\mathbf{x}}(\cdot; \theta)$, then for any one-one and onto function $h = g(\theta)$, the MLE of h is given by $h' = g(\theta')$.
- **Proof:**
 - 1) h' is the MLE of the parameter h , so $h' = \operatorname{argmax}_h L(h|\mathbf{x})$ where $\mathbf{x}$ = set of available samples of X . Here L is the log likelihood function.
 - 2) Hence $h' = \operatorname{argmax}_h (\max_{\theta: g(\theta)=h} L(\theta|\mathbf{x}))$.
 - 3) But $\max_h (\max_{\theta: g(\theta)=h} L(\theta|\mathbf{x})) = \max_{\theta: g(\theta)=h} (\max_h L(\theta|\mathbf{x})) = \max_{\theta} L(\theta|\mathbf{x})$.
 - 4) Note that $\theta' = \operatorname{argmax}_{\theta} L(\theta|\mathbf{x})$
 - 5) We now proceed by contradiction.
 - a) Now suppose that $\theta' \notin g^{-1}(h')$.
 - b) Then $\max_{\theta: g(\theta)=h'} L(\theta|\mathbf{x}) < \max_{\theta} L(\theta|\mathbf{x}) = L(\theta'|\mathbf{x})$ which contradicts step 3.
 - 6) Hence we must have $\theta' \in g^{-1}(h')$ and hence $h' = g(\theta')$.

Functional Invariance of MLEs

- **Invariance theorem for MLEs:** If θ' is the MLE of the parameter θ of a density function $f_x(\cdot; \theta)$, then for any one-one and onto function $\eta = g(\theta)$, the MLE of η is given by $\eta' = g(\theta')$.
- **Proof:**
 - 1) Let g be a one-to-one function. So we have:

$$\max L(g(\theta)) = \max_{\eta=g(\theta)} L(\eta) = \max_{\eta} L(g^{-1}(\eta))$$

But the maximum occurs when

$$g^{-1}(\eta) = \hat{\theta} \longrightarrow \hat{\eta} = g(\hat{\theta})$$

Functional Invariance of MLEs

- What happens when g is not a one-to-one function?

$$g^{-1}(\eta) = \{\theta : g(\theta) = \eta\}$$

There can be many θ values that give rise to any η .

Therefore there can be many θ values such that $g(\theta) = \eta'$ which maximizes the likelihood.

This maximum value of the likelihood is the same as $L(\theta')$.

This gives us $\eta' = g(\theta')$.

Functional invariance of MLE

- Consider that we want to find the standard deviation σ of a Gaussian distribution from its samples.
- We can set the derivative of the log likelihood w.r.t. σ to zero to find the MLE for σ , i.e. σ' .
- Alternatively, consider the variance $v = \sigma^2$.
- Rewrite the log likelihood in terms of v and set its derivative w.r.t. v to zero, to obtain the MLE v' .
- Then obtain σ' as the positive square root of v' .

Bayesian Estimation

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Bayesian Estimation

- In maximum likelihood estimation, we regarded the parameter θ of $f_x(x;\theta)$ as a fixed value.
- In Bayesian estimation, we regard θ as a random variable with some **prior distribution** denoted by $p(\theta)$.
- The prior records our earlier knowledge about θ (eg: given to us by a domain expert).
- Eg: If we believe that θ lies somewhere between 0 and 1, we could assume a $\text{Uniform}(0,1)$ prior on θ .
- Eg: If we believe that θ has a strong chance of being equal to 1 and rapidly decreasing chances of being on either side of 1, we can use a prior of $N(1, \sigma^2)$.

Bayesian Estimation

- Consider that we have iid samples $X_1, X_2, \dots, X_n$ from $f(x|\theta)$.
- Note that the prior distribution is not influenced by these samples.
- The likelihood of these samples given θ is $f(\mathbf{x}|\theta)$.
- Now, we consider the conditional PDF of θ given $\mathbf{x}$ – called the **posterior probability density** –as follows:

$$\begin{aligned} f(\theta|x_1, \dots, x_n) &= \frac{f(\theta, x_1, \dots, x_n)}{f(x_1, \dots, x_n)} \\ &= \frac{p(\theta)f(x_1, \dots, x_n|\theta)}{\int f(x_1, \dots, x_n|\theta)p(\theta) d\theta} \end{aligned}$$

Bayesian Estimation

- There are two ways of obtaining θ now.
- In one method, we find θ which maximizes the posterior PDF.
- This is called the MAP or maximum a posteriori method.
- Maximum likelihood estimation is equivalent to MAP if we assumed a uniform prior on θ .

$$\begin{aligned} f(\theta|x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n|\theta)p(\theta)}{\int_a^b f(x_1, \dots, x_n|\theta)p(\theta) d\theta} \\ &= \frac{f(x_1, \dots, x_n|\theta)}{\int_a^b f(x_1, \dots, x_n|\theta) d\theta} \end{aligned} \quad a < \theta < b$$

$$\theta \sim \text{Uniform}(a,b)$$

Bayes Estimator

- In the second method, which we call Bayes estimator, we seek to find θ which minimizes the mean squared error given the data $\{X_i\}$ as well as the prior $p(\theta)$.
- We know that the “mean” minimizes the MSE, so we obtain the mean of the posterior distribution. Thus the Bayes estimator is:

$$E[\theta|X_1 = x_1, \dots, X_n = x_n] = \int \theta f(\theta|x_1, \dots, x_n) d\theta$$

Example 1

- Consider n iid Bernoulli(θ) samples where the prior on θ is Uniform(0,1).
- Find the Bayes estimator of θ .

$$p(\theta) = 1, \quad 0 < \theta < 1$$

$$\begin{aligned} f(\theta|x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n, \theta)}{f(x_1, \dots, x_n)} \\ &= \frac{f(x_1, \dots, x_n|\theta)p(\theta)}{\int_0^1 f(x_1, \dots, x_n|\theta)p(\theta) d\theta} \longrightarrow \\ &= \frac{\theta^{\sum_1^n x_i} (1 - \theta)^{n - \sum_1^n x_i}}{\int_0^1 \theta^{\sum_1^n x_i} (1 - \theta)^{n - \sum_1^n x_i} d\theta} \end{aligned}$$

Recall that we are using **Bayes rule** here:
 $P(A|B)P(B) = P(B|A)P(A) = P(A, B)$ for any random variables A and B

$$\int_0^1 \theta^m (1 - \theta)^r d\theta = \frac{m!r!}{(m + r + 1)!}$$

$$x = \sum_{i=1}^n x_i$$

$$f(\theta|x_1, \dots, x_n) = \frac{(n + 1)! \theta^x (1 - \theta)^{n-x}}{x! (n - x)!}$$

For $n = 10$ trials with success probability θ , suppose we have 6 successes. The ML estimate for $\theta = 6/10 = 0.6$. The Bayes estimate is $(6+1)/(10+2) = 7/12$.

This is called the **beta distribution** with parameters n and x

$$\begin{aligned} E[\theta|x_1, \dots, x_n] &= \frac{(n + 1)!}{x!(n - x)!} \int_0^1 \theta^{1+x} (1 - \theta)^{n-x} d\theta \\ &= \frac{(n + 1)!}{x!(n - x)!} \frac{(1 + x)!(n - x)!}{(n + 2)!} = \frac{x + 1}{n + 2} = \frac{\sum_{i=1}^n X_i + 1}{n + 2} \end{aligned}$$

What is the MAP here?

We need to find the mode of the following posterior density:

$$f(\theta|x_1, \dots, x_n) = \frac{(n+1)! \theta^x (1-\theta)^{n-x}}{x! (n-x)!}$$

Setting the derivative w.r.t. θ to 0, we get:

$$x\theta^{x-1}(1-\theta)^{n-x} + \theta^x(1-\theta)^{n-x-1}(n-x)(-1) = 0$$
$$\theta = \frac{x}{n}$$

This is the same as the MLE as we have a uniform prior!

Example 2: Mean of Gaussian distribution with known variance

$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ^2

Prior: $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$

$$L(\mu) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right) = \exp \left(-\frac{n(\bar{X} - \mu)^2}{2\sigma^2} \right) \times (\text{const})$$

$$p(\mu \mid X_{1:n}) = \mathcal{N}(\mu_{\text{post}}, \sigma_{\text{post}}^2)$$

We will do detailed derivations for these steps in the next 2-3 slides.

$$\mu_{\text{post}} = \frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}} = \frac{n\sigma_0^2 \bar{X} + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2} \quad \sigma_{\text{post}}^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}} = \frac{\sigma^2 \sigma_0^2}{n\sigma_0^2 + \sigma^2}$$

$$L(\mu) = p(X_1, \dots, X_n \mid \mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right)$$

Likelihood function

$$L(\mu) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right)$$

Likelihood function with constant factors ignored

$$p(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) \propto \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$

Prior

$$p(\mu \mid X_1, \dots, X_n) \propto L(\mu) p(\mu) \propto \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 - \frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right]$$

The posterior via Bayes rule (ignoring constant factors as we will ignore them in the MAP)

$$p(\mu \mid X_1, \dots, X_n) \propto L(\mu) p(\mu) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 - \frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right]$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

Independent of μ

$$p(\mu \mid X_1, \dots, X_n) \propto \exp \left[-\frac{n(\bar{X} - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right]$$

$$p(\mu \mid X_1, \dots, X_n) \propto \exp \left[-\frac{n(\bar{X} - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right]$$

$$-\frac{1}{2} \left[\frac{n}{\sigma^2} (\mu^2 - 2\bar{X}\mu) + \frac{1}{\sigma_0^2} (\mu^2 - 2\mu_0\mu) \right] + (\text{terms not involving } \mu)$$

$$-\frac{1}{2} \left[\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \mu^2 - 2 \left(\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right) \mu \right] + \text{const}$$

$$\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \left(\mu - \frac{\frac{n\bar{X}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}} \right)^2 + \text{const.}$$

Complete the square: the terms introduced in square completion are independent of μ and can be ignored.

$$p(\mu \mid X_1, \dots, X_n) = \frac{1}{\sqrt{2\pi\sigma_{\text{post}}^2}} \exp\left[-\frac{(\mu - \mu_{\text{post}})^2}{2\sigma_{\text{post}}^2}\right]$$

$$\mu_{\text{post}} = \frac{n\sigma_0^2 \bar{X} + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}, \quad \sigma_{\text{post}}^2 = \frac{\sigma^2 \sigma_0^2}{n\sigma_0^2 + \sigma^2}$$

$$\text{Posterior precision} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}$$

- As the posterior is Gaussian, its mode is equal to the mean. So the MAP for μ is equal to μ_{post} .
- The Bayes estimate under the squared loss is the mean which in this case equals the MAP.

The posterior mean is a weighted combination of the sample mean $\bar{X}$ and the prior mean μ_0

$$\mu_{\text{post}} = w\bar{X} + (1 - w)\mu_0 \qquad w = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2}$$

- If the prior variance $(\sigma_0)^2 \rightarrow \infty$, then $w \rightarrow 1$ and the posterior mean tends to the sample mean.
 - Here you trust only the data and the prior is **non-informative** (not surprising, as the prior is nearly uniform)
- If $n = 0$ (no data), then $w = 0$ and you can trust only the prior.
 - In particular, Bayesian methods are very useful if n is very small.
- If $n \rightarrow \infty$, again w tends to 1, you trust only the data and the prior does not have much use.
- If $(\sigma_0)^2 \rightarrow 0$, i.e. you have peaky priors, then $w = 0$ and you trust only the prior and not the data.

Concept of conjugate priors

- In the earlier example, the likelihood $f(x|\theta)$ is Gaussian.
- For this likelihood, the posterior $f(\theta|x)$ is Gaussian if the prior $p(\theta)$ is Gaussian, albeit with different parameters.
- Given a likelihood $f(x|\theta)$, if the posterior probability (density) $f(\theta|x)$ is in the same family as the prior probability (density) $p(\theta)$, then we say that the prior and posterior are **conjugate** to each other.
- The prior in such cases is called a **conjugate prior** for the likelihood $f(x|\theta)$.
- If the likelihood $f(x|\theta)$ changes, the prior $p(\theta)$ may no longer be conjugate.
- The conjugate prior is mostly an algebraic convenience as it gives a closed-form formula for the posterior.
- In the absence of a closed-form expression, we would need to do numerical integration to compute the posterior.

Bayesian estimation for success parameter of Bernoulli

$$X_i \sim \text{Bernoulli}(p) \quad P(X_i = 1 \mid p) = p, \quad P(X_i = 0 \mid p) = 1 - p$$

$$k = \sum_{i=1}^n X_i \quad \text{Total number of successes}$$

$$p \sim \text{Beta}(\alpha, \beta) \quad \text{Prior for the success parameter } p$$

$$\pi(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}, \quad 0 < p < 1$$

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)} \quad \Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$$

The likelihood function is as follows:

$$L(p) = P(X_1, \dots, X_n \mid p) = p^k (1 - p)^{n-k}$$

$$\pi(p \mid X_1, \dots, X_n) \propto L(p) \pi(p)$$

Posterior density

$$\pi(p \mid \text{data}) \propto p^k (1 - p)^{n-k} \times p^{\alpha-1} (1 - p)^{\beta-1} = p^{\alpha+k-1} (1 - p)^{\beta+n-k-1}$$

$$p \mid \text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

$$\pi(p \mid \text{data}) = \frac{1}{B(\alpha + k, \beta + n - k)} p^{\alpha+k-1} (1 - p)^{\beta+n-k-1}$$

- To find the MAP estimate of p , we will maximize the posterior density given above (rather the logarithm of the posterior density) w.r.t. P
- Note that the posterior is again a Beta density, quite like the prior. So this is one more example of a conjugate prior.

$$\ell(p) := \log \pi(p \mid \text{data}) = (\alpha + k - 1) \log p + (\beta + n - k - 1) \log(1 - p) + C$$

$$\frac{d\ell}{dp} = \frac{\alpha + k - 1}{p} - \frac{\beta + n - k - 1}{1 - p} = 0$$

$$(\alpha + k - 1)(1 - p) = (\beta + n - k - 1)p$$

$$\Rightarrow (\alpha + k - 1) = (\alpha + \beta + n - 2)p$$

$$\hat{p}_{\text{MAP}} = \frac{\alpha + k - 1}{\alpha + \beta + n - 2}$$

Contrast this with the MLE estimate of p , which is k/n .

We will now derive the Bayes estimate of p .

$$p \mid \text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

$$\hat{p}_{\text{Bayes}} = \mathbb{E}[p \mid \text{data}] = \int_0^1 p \pi(p \mid \text{data}) dp$$

$$\pi(p \mid \text{data}) = \frac{1}{B(\alpha + k, \beta + n - k)} p^{\alpha+k-1} (1 - p)^{\beta+n-k-1}$$

$$\mathbb{E}[p \mid \text{data}] = \frac{1}{B(\alpha + k, \beta + n - k)} \int_0^1 p p^{\alpha+k-1} (1 - p)^{\beta+n-k-1} dp$$

$$= \frac{1}{B(\alpha + k, \beta + n - k)} \int_0^1 p^{(\alpha+k)+0} (1 - p)^{\beta+n-k-1} dp$$

$$\begin{aligned}
\mathbb{E}[p \mid \text{data}] &= \frac{1}{B(\alpha + k, \beta + n - k)} \int_0^1 p p^{\alpha+k-1} (1-p)^{\beta+n-k-1} dp \\
&= \frac{1}{B(\alpha + k, \beta + n - k)} \int_0^1 p^{(\alpha+k)+0} (1-p)^{\beta+n-k-1} dp \\
&= \frac{1}{B(\alpha + k, \beta + n - k)} B(\alpha + k + 1, \beta + n - k)
\end{aligned}$$

$$B(u, v) = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}$$

$$\mathbb{E}[p \mid \text{data}] = \frac{\Gamma(\alpha + k + 1)\Gamma(\beta + n - k)}{\Gamma(\alpha + \beta + n + 1)} \cdot \frac{\Gamma(\alpha + \beta + n)}{\Gamma(\alpha + k)\Gamma(\beta + n - k)}$$

$$\begin{aligned}
\mathbb{E}[p \mid \text{data}] &= \frac{\Gamma(\alpha + k + 1)\Gamma(\beta + n - k)}{\Gamma(\alpha + \beta + n + 1)} \cdot \frac{\Gamma(\alpha + \beta + n)}{\Gamma(\alpha + k)\Gamma(\beta + n - k)} \\
&= \frac{\Gamma(\alpha + k + 1)}{\Gamma(\alpha + k)} \cdot \frac{\Gamma(\alpha + \beta + n)}{\Gamma(\alpha + \beta + n + 1)} \\
&= \frac{\alpha + k}{\alpha + \beta + n}
\end{aligned}$$